Tejaswi Samavedula

@ tejaswisamavedula@gmail.com | in/tejaswisamavedula | tejaswisam | Portfolio

Education

Indian Institute of Information Technology Design and Manufacturing — Chennai, India
B.Tech in Smart Manufacturing; **CGPA: 8.47/10.00** — *Dec 2021 – July 2025*

Relevant coursework: Robotics and Automation, Analysis and Synthesis of Robot Mechanisms, Linear Algebra, Machine-to-Machine Communication, Sensors and Control, Embedded Systems, Prototyping and Testing, Production Drawing Practice, Operations Research

Experience

Graduate Engineer Trainee - Robotics and Automation — Hyderabad, India
Robotspace Robotics and Automation — *Aug 2025 – Present, Full-time*

- Developed a PX4-based outdoor AMR for Hyundai, leading end-to-end development for autonomous lux-data collection.
- Integrated industrial PLCs, AMRs, and robotic systems.
- Researched autonomous robotic welding using the Yaskawa AR1440 with ROS 2 and MoveIt.

Mechatronics Engineer Intern — Hyderabad, India
Techolution India — *Jan 2025 – June 2025, Internship*

- Collaborated with the electronics team on PCB design, sensor positioning, and wiring optimization.
- Contributed to firmware development, assisting in integration of sensor data streams.
- Generated URDF files for the Robotic AI Hand Assembly
- Designed mechanical components and mechanisms for a Robotics AI Hand, ensuring functionality, manufacturability, and seamless integration with electronics.

Research Intern — Chennai, India
Autonomous Robotics (ARISE) Lab, IIITDM Kancheepuram — *May 2024 – Dec 2024, Internship*

- Conducted research on indoor mapping and navigation using nano-drones, focusing on efficient and low-cost autonomous exploration
- Developed and tested algorithm for navigation in GPS-denied environments using ROS2 and tested in gazebo simulation and verifed through hardware implementation
- Demonstrated the research at COMSNETS 2025 Demos and Exhibits
- Proceedings were published in IEEE Xplore | *View Publication*

Team Captain — Chennai, India
Astra IIITDM - Aerial Robotics Team — *Jan 2024 – Dec 2024, Part-time*

- Led a 24-member interdisciplinary aerial robotics team that participated in SAE AeroTHON 2024.
- Contributed to the development of an autonomous navigation stack for a quadcopter drone using ROS2 and PX4 flight stack.

Projects

Indoor Obstacle Avoidance and for Drones | *Crazyflie, Python, ROS2, Gazebo* | GitHub

- Developed ROS2 based Indoor Obstacle Avoidance and Navigation ROS2 algorithm for Crazyflie nano-drones using crazyswarm2 library and Multiranger sensor. Simulated in Gazebo environment and then implemented on hardware.

Autonomous Docking with TurtleBot3 in ROS 2 | *Python, ROS2, Nav2, Gazebo* | GitHub

- Developed a autonomous robot navigation and docking using ArUco markers and a TurtleBot3 in a Gazebo simulation, built using ROS 2 Humble. Navigates through obstacles using Nav2

Design and Development of TRACE Device | *C++, CAD Modeling, ESP32, PlatformIO, FreeRTOS* | GitHub

- Developed TRACE (Tool for Rapid Analysis, Calibration and Evaluation) Device (v1), a custom-built, handheld multi-sensor system for capturing orientation, distance, and directional data in real-time.

Skills

Languages/Frameworks: C/C++, Python, ROS2

Areas: CAD, PCB Designing

Developer tools: Git, Docker, Bash, Github

Software: VSCode, PlatformIO, SolidWorks (CSWA)

Hardware: ESP32, STM32, Arduino